Diego F. Rojas-Torres

Biomedical Engineer · Health Systems Researcher · Digital Health Leader

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PROFILE

Biomedical engineer and PhD candidate in Engineering and Sciences with the Industry at Pontifical Catholic University of Chile, with experience across healthcare systems, digital health, clinical engineering, applied research, and institutional transformation. Currently Researcher and Head of Digital Transformation at Chile's National Cancer Institute and Visiting Researcher in the Department of Industrial and Operations Engineering at the University of Michigan. His work connects quantitative methods with clinical and organizational realities, with interests in oncology care pathways, patient prioritization, process mining, discrete-event simulation, and data-driven decision support.

RESEARCH AREAS

- Healthcare systems research and operational research
- Oncology care pathways and patient prioritization
- Process mining and discrete-event simulation
- Digital transformation and health information systems
- Leadership of interdisciplinary clinical and technical teams
- Academic, clinical, and institutional collaboration
- Health data science and decision-support development
- Technology implementation, training, and knowledge translation

CURRENT RESEARCH AND AFFILIATIONS

Doctoral research. Simulation-Based Patient Management and Data- and Process-Mining-Driven Waiting-List Administration: An Applied Oncology Case Study. Original Spanish title: Modelo de gestión de pacientes mediante simulación y administración de listas de espera basada en minería de datos y procesos: caso aplicado del paciente oncológico. Advisor PhD. Alejandro Mac Cawley.

Visiting Researcher. Department of Industrial and Operations Engineering, University of Michigan, Ann Arbor, MI, USA. August 3, 2026 – December 18, 2026. Working with Professor PhD. Mariel S. Lavieri.

International collaborations. Research collaborations involving the University of Michigan (US), Université Laval (Canada), Pontificia Universidad Católica de Chile, and the National Cancer Institute of Chile.

PROFESSIONAL EXPERIENCE

Researcher and Head of Digital Transformation (ex-Head of IT)
National Cancer Institute of Chile, Santiago, Chile (2023–present)
Leads and coordinates digital transformation initiatives in a national oncology institution. Works across clinical, technical, and management teams to support health information systems, data use, institutional coordination, and technology-enabled improvement. Currently leading a 10-person team composed of developers, engineers, technicians, and clinicians.

Research and Applied Technology Collaborator
Plataforma 360 Ltda., Healthcare Technology Innovation Consultancy (2019–present)
Participates in applied research, healthcare innovation, process improvement, and the design and implementation of technology-based solutions for healthcare organizations.

Head of Digital Health and Innovation
Servicio de Salud Metropolitano Sur, Chile (2017–2023)
Managed and coordinated digital health, innovation, and health information initiatives within the public healthcare system.

Clinical Engineer, Critical Care Equipment
Hospital Clínico Universidad de Chile (2016–2017)
Provided clinical and technical management of critical medical equipment and coordinated with healthcare teams.

Research Assistant
Universidad de Valparaíso, Regional Telemedicine Project (2015)
Conducted field assessments and needs analysis for telemedicine and primary-care initiatives.

SELECTED PROJECTS

National TeleStroke Network in Chile. Technology lead and implementation coordinator for a centralized Hub-and-Spoke telemedicine system across 10 public hospitals. Funded by Chile's Ministry of Health and Servicio de Salud Metropolitano Sur (2017–2020).

TeleTriage, a primary-care demand management software. External consultant during the initial pilot stages of a digital demand-management system, contributing healthcare process optimization and health information systems expertise. Pan American Health Organization project (2020–2021).

e-Move App. Gamified software for home-based rehabilitation of distal radius fractures, developed at Hospital Clínico Mutual de Seguridad (Santiago, Chile) and funded by the Superintendencia de Seguridad Social (SUSESO), Chile. Served as Technical Director in a randomized clinical trial, leading biomechanical image recognition and processing systems and coordinating clinical–technical integration (2019–2022).

PreD System. Digital IoT hardware/software system for delirium prevention, developed at Hospital Clínico Mutual de Seguridad (Santiago, Chile) and funded by the Superintendencia de Seguridad Social (SUSESO), Chile. Consultant in the design, prototyping, implementation, and staff training for environmental monitoring and communication systems for hospitalized patients (2020–2023).

EDUCATION

PhD Candidate in Engineering and Sciences with the Industry, Pontificia Universidad Católica de Chile (2021–present)
Postgraduate Diploma in Healthcare Institution Management, Universidad de Santiago de Chile (2018–2019)
Civil Biomedical Engineer, Universidad de Valparaíso (2010–2016)

PUBLICATIONS AND MANUSCRIPTS

Manuscript under peer review. Rojas-Torres, D., Mac Cawley, A., Ruiz, A., Lavieri, M. S., Abarca, I., & Flores, F. Prioritizing Access to Cancer Treatment in Low-Resource Countries: A Systematic Review. Public Health Reviews. Under peer review after passing initial editorial screening. Collaborative work with Université Laval and University of Michigan. PROSPERO: CRD420261334566.

Manuscript in preparation. A Conceptual Framework for a Prognostic Decision-Support Tool in Oncology Treatment Prioritization under Resource Constraints: Case Illustrations from a Tertiary Cancer Center.

Manuscript in preparation. Redesigning Cancer Patient Care by Linking Oncology Care Pathways, Resource Capacity, and Patient Outcomes: A Process Mining and Simulation Case Study at a High-Complexity Cancer Center.

González, X., Cerda, B., Ciudad, A. M., Abarca, I., **Rojas-Torres, D.**, et al. (2025). Cancer research in Chile: beyond universities, contributions from the National Cancer Institute. International Journal of Medical and Surgical Sciences, 12(1), 1–4. DOI: 10.32457/ijmss.v12i1.2997.

Mansilla, E., Mazzon, E., Cárcamo, D., Jurado, F., Lara, L., **Rojas-Torres, D.**, et al. (2019). Telestroke in Chile: 1 year experience at 7 hospitals. Revista Médica de Chile, 147(9), 1107–1113. DOI: 10.4067/S0034-98872019000901107.

Mazzon, E., Delfino, C., Mirelis, S., **Rojas-Torres, D.**, et al. (2022). Breaking Down Barriers: Easter Island's First Telestroke Thrombolysis Experience and Case Report. Journal of Stroke and Cerebrovascular Diseases, 31(6), 106412. DOI: 10.1016/j.jstrokecerebrovasdis.2022.106412. PMID: 35354109.

PRESENTATIONS AND CONFERENCE CONTRIBUTIONS

- Transparent Data-Driven Oncology Prioritization: A Pathway-Specific Proof of Concept. Poster, 2026 SIAM Student Mini-Symposium in Applied Mathematics, University of Michigan, August 29, 2026. Sponsored by MCAIM.
- Data-Driven Cancer Patient Waiting List Management. HOPER 2025 School on Health Operations Research, Université du Québec à Montréal.
- Advanced Management of Oncology Waiting Lists Using Prospective Methods of Prioritization. Biomedical Engineering Colloquium Series, University of Valparaíso (2025).
- Information Technologies and Systems Required for a Basic TeleStroke Implementation in a Public Healthcare Center in Chile. LXXIV Congress of the Chilean Society of Neurology, Psychiatry, and Neurosurgery (2019).

- Analysis of a Provincial Health Network in Northern Chile During Disaster and Emergency Situations. XXXII Chilean Public Health Conference, University of Chile, Santiago (2015).

- Gaps and Future Perspectives in the Use of Technology During Disaster and Emergency Situations Caused by Flood Events in Northern Chile. IEEE EMBS International Biomedical Engineering Student Conference & VIII Annual Biomedical Engineering Congress, University of Concepción (2015).

TEACHING AND MENTORING

- Guest Lecturer, Medical Informatics — Universidad del Desarrollo, 2023–2024. Teaching in the Health Processes and Information Systems module for undergraduate Medical Technology students.

- Collaborative Learning Program Mentor — Universidad de Valparaíso, 2015. Academic support in mathematics, physics, and biology; student follow-up and coordination with teaching and support units.

AWARDS AND FUNDING

- ANID National Doctoral Scholarship, Chile (2022).

- IFORS–ALIO Travel Scholarship for HOPER 2025.

METHODS AND TECHNICAL SKILLS

Process mining · discrete-event simulation · optimization models · machine learning · health data science · survival analysis · biomedical technologies · project and team management / leadership skills.

Python · R · SQL · MATLAB · C++ · HL7/FHIR interoperability · DevOps skills (PHP, Java, CSS & HTML)

ACADEMIC PROFILES

ORCID: https://orcid.org/0009-0008-2136-9110
Google Scholar: https://scholar.google.com/citations?user=VbLf-CYAAAAJ&hl=es
ResearchGate: https://www.researchgate.net/profile/Diego-Rojas-54
GitHub: https://github.com/diegofrt
LinkedIn: https://www.linkedin.com/in/drojast/